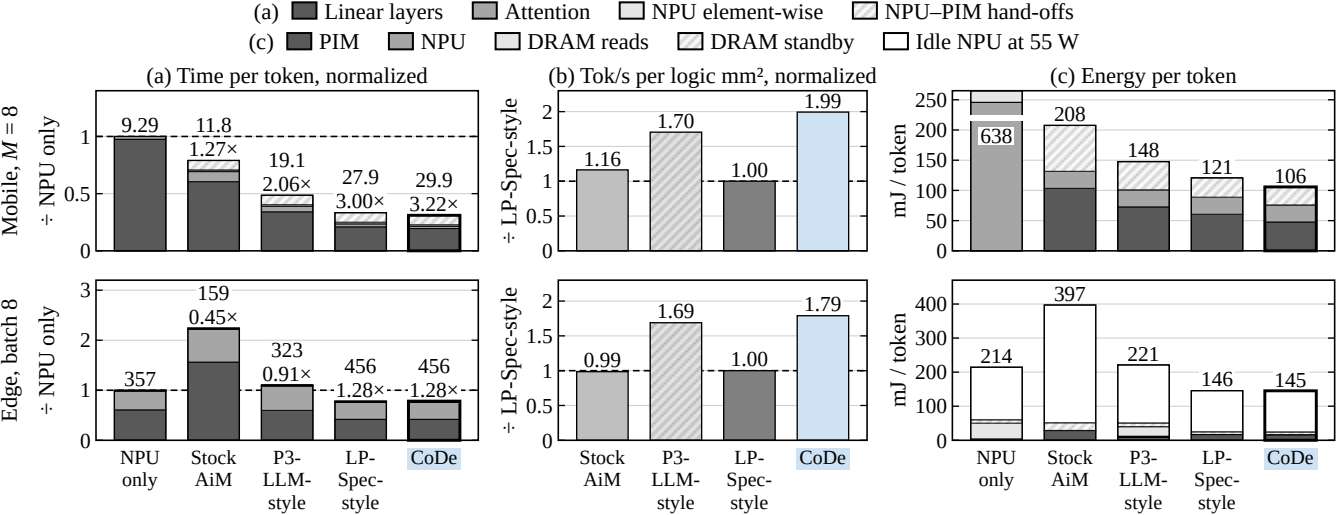

(a) Linear layers
Attention
NPU element-wise
NPU–PIM hand-offs
(c) PIM
NPU
DRAM reads
DRAM standby
Idle NPU at 55 W
(a) Time per token, normalized
(b) Tok/s per logic mm², normalized
(c) Energy per token
Mobile, M = 8
Edge, batch 8
÷ NPU only
÷ LP-Spec-style
mJ / token
9.29
11.8
1.27×
19.1
2.06×
27.9
3.00×
29.9
3.22×
1.16
1.70
1.00
1.99
638
208
148
121
106
357
159
0.45×
323
0.91×
456
1.28×
456
1.28×
0.99
1.69
1.00
1.79
214
397
221
146
145
NPU only
Stock AiM
P3-LLM-style
LP-Spec-style
CoDe